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Foreword

This Technical Report (TR) has been produced by ETSI Technical Committee Securing Artificial Intelligence (SAI).

Modal verbs terminology

In the present document "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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Introduction

Machine learning use cases in telecom have shown great potential in assisting with solving problems such as anomaly detection, root cause analysis, managed services, and network optimization. The present document aims to study AI security and privacy aspects that are specific to the telecom industry use cases.

1 Scope

The use of AI to facilitate the use cases may cause AI security and privacy issues specific to the telecom industry. The scope of this proposed work item will be to investigate security and privacy issues related to the use of AI in the telecom industry sector. Harmonisation with 3GPP work in SA1, SA2, and SA3 is anticipated.

Key AI use cases in telecom networks are (non-exhaustive list):

- Network as a service.
- Network optimization.
- Network planning and upgrades.
- Automating security operations (anomaly detection, planning mitigation and response).

This investigation may use but is not limited to the Network Operations Lifecycle Phases methodology.

2 References

2.1 Normative references

Normative references are not applicable in the present document.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents may be useful in implementing an ETSI deliverable or add to the reader's understanding, but are not required for conformance to the present document.

[i.1] 3GPP TS 23.228: "Architecture enhancements for 5G System (5GS) to support network data analytics services (Release 18)".

[i.2] 3GPP TS 28.104: "Management and orchestration; Management Data Analytics (MDA) (Release 17)".

[i.3] 3GPP TR 37.817: "Study on enhancement for data collection for NR and ENDC".

[i.4] ETSI TS 138 423: "5G; NG-RAN; Xn Application Protocol (XnAP) (3GPP TS 38.423 version 18.5.0 Release 18)".

[i.5] Open Source MANO (OSM) - ETSI.

[i.6] Planning for Network Day 0, 1, and 2 Tasks and Stumbling Blocks.

[i.7] The difference between day-0, day-1, and day-2 operations.

[i.8] Day 0, Day 1, Day 2 Operations: Putting it All Together on Day 2.

[i.9] ETSI GR NFV-EVE 022: "Network Functions Virtualisation (NFV) Release 5; Architectural Framework; Report on VNF configuration".

[i.10] Pialla, G., Ismail Fawaz, H., Devanne, M. et al.: "Time series adversarial attacks: an investigation of smooth perturbations and defense approaches". Int J Data Sci Anal (2023).

[i.11] Siyang Lu, Mingquan Wang, Dongdong Wang, Xiang Wei, Sizhe Xiao, Zhiwei Wang, Ningning Han, Liqiang Wang: "Black-box attacks against log anomaly detection with adversarial examples", Information Sciences, Volume 619, 2023, Pages 249-262, ISSN 0020-0255.

3 Definition of terms, symbols and abbreviations

3.1 Terms

3.1.1 Network Operations Lifecycle Phases

The present document adopts a commonly used nomenclature used for describing activities pertaining to the commissioning of a service, namely Day 0 - Day N. This approach and terms have been used in the ETSI Open Source MANO [i.5] project and their application explained in [i.6], [i.7], and [i.8] to describe the onboarding of Containerized Network Functions, with some concrete use cases to be found in [i.9]. For the scope of the present document, the terms have been redefined as below.

Day 0

The planning and evaluation phase for a new telecommunications network encompassing all decision-making processes prior to deployment activities.

Day 1

The deployment phase for a new network, spanning all activities necessary for the commissioning of infrastructure, systems, and supply chains required for network operations.

Day 2

The steady-state operations and maintenance phase for a network, post initial installation, until the final decommissioning of the network.

3.1.2 NIST AI Attack Taxonomy

AI supply chain: manipulation of training data or AI/ML model or AI/ML supporting software libraries

evasion: manipulating data which results in misclassification or no detection

poisoning : manipulating training data which results in model learning incorrectly

privacy: extracting sensitive information model was trained on

prompt injection: submission of malicious prompts to an AI system either directly or through ingestion sources

3.1.3 NIST AI Attacker Goals

abuse violation: abuse of a deployed AI/ML model to achieve attacker goals

availability breakdown: degradation of AI/ML model performance during deployment

integrity violation: erosion of model integrity to elicit incorrect results either through evasion or poisoning

privacy compromise: discovery of information pertaining to training data or model characteristics

3.2 Symbols

Void.

3.3 Abbreviations

ADRF	Analytics Data Repository Function
AF	Application Function
AI	Artificial Intelligence
AI/ML AF	AI/ML Application Function
AKMA	Authentication and Key Management for Applications
AMF	Access Management Function
AN	Access Network
AnLF	Analytics Logical Function
AS	Access Stratum
CI/CD	Continuous Integration and Continuous Delivery
CN	Core Network
CoA	Course of Action
Cp	Control Plane
CP	Control Plane
CPU	Central Processing Unit
DCCF	Data Collection Coordination Function
DDoS	Distributed Denial of Service
E2E	End-to-End
GBA	Generic Bootstrapping Architecture
GenAI	Generative AI
GPU	Graphic Processing Unit
IaC	Infrastructure as Code
IN	Intelligent Network
KPI	Key Performance Indicator
LLM	Large Language Model
MANO	Management And Orchestration
MDA	Management Data Analytics
MDAS	Management Data Analytics Service
MDT	Minimization of Drive Test
MFAF	Messaging Framework Adaptor Function
ML	Machine Learning
MTLF	Model Training logical function
NAS	Non-Access Stratum
NEF	Network Exposure Function
NF	Network Function
NN	Neural Network
NOC	Network Operations Centre
NRF	Network Repository Function
NSACF	Network Slice Admission Control Function
NWDAF	NetWork Data Analytics Function
OAM	Operations, Administration and Maintenance
PCA	Principal Component Analysis
PCF	Policy Control Function
PDCP	Packet Data Convergence Protocol
QoE	Quality of Experience
QoS	Quality of Service
RAG	Retrieval Augmented Generation
RAN	Radio Access Network
RAT	Radio Access Technology
RCA	Root Cause Analysis
RCEF	RRC Connection Establishment Failure
RLF	Radio Link Failure
RRC	Radio Resource Control
SBI	Service Based Interface
SIP	Session Initiation Protocol
SLA	Service Level Agreement
SMART	Self-Monitoring, Analysis, and Reporting Technology
SMF	Session Management Function
SOC	Security Operations Centre

t-SNE	t-distributed Stochastic Neighbour Embedding
UDM	Unified Data Management
UDR	Unified Data Repository
UE	User Equipment
UP	User Plane

4 Convention Description

4.1 Notation

For the purpose of the present document, the following notations apply:

<information> stands for the "information" been exchanged or transmitted between different modules or via interfaces.

[security component] stands for the "security components" described in clause 7 that has involved in the interactive procedures.

(optional) means the description followed by is the alternative that has enhanced effect but more strict requirements compared. It may be referred to specific situations when implementing a certain security component or a security mechanism.

5 Overview

5.1 Use of Generative AI vs. traditional AI in telecom providers' networks

In many cases, traditional AI (as opposed to Generative AI (GenAI)) might be sufficient in telecom providers' networks.

Many of the use cases described in further clauses (e.g. Anomaly detection, Customer churn prediction, Predictive maintenance, Root cause identification, System monitoring, Ticket classification, and routing) may be supported by traditional AI.

While GenAI may be suitable for interactive experiences, such as customer care and postmortem assessment of a network error, a lot of other optimization efforts - such as initial detection of errors and root cause analysis - might not need more than traditional ML models.

However, when, for example, traditional AI detects a new cyber threat and new firewall rules need to be written to address it or a new threat signature is discovered for future use by traditional AI, GenAI may be employed to generate and deploy these rules and/or signatures.

5.2 ML functionality in telecom providers' networks

5.2.1 Data collection and preparation mechanisms

In the context of 5G / 3GPP-enabled standards, data collection forms part of the core functionality of the Management Data Analytics (MDA) capability of the network. More generally, the telecom provider should ensure the availability of structured, real-time operational and performance characteristics of their network, such as QoS/QoE metrics, throughput, and measured network latency. Additionally, unstructured data sources such as natural-language trouble tickets generated increasingly represent viable sources of information for AI models, especially in conjunction with the aforementioned structured information.

Depending on the nature of the AI/ML models employed, several pre-processing steps may be required. Structured, quantitative data may require statistical transformations such as normalization to contain their co-domain. In cases where a large number of features (i.e. input parameters) are available, dimensionality reduction techniques such as Principal Component Analysis (PCA), or t-distributed Stochastic Neighbour Embedding (t-SNE) may be employed to reduce the feature space. Natural language data typically undergoes a string of techniques such as normalization (e.g. non-printed character removal or white space normalization), deduplication, and - depending on the use case - removal of private data, pseudonymization, or anonymization for security and privacy compliance. The following steps may include lemmatization, grouping related words into singular concepts (lemmas), and vectorization, which is the conversion of these lemmas to mathematical vector representations. This subsequently allows for sentiment, syntax, and semantic analysis, to determine the premise of a given text. The interfaces provided by Large Language Models (LLMs) typically perform these steps transparently.

5.2.2 Model engineering and evaluation mechanisms

Model engineering refers to the process of designing ML "pipelines" that convert raw data into actionable inferences by performing a set of transformations. This process is executed based on the insights of domain experts. In the context of a telecommunications network, this translates to the Network Operations Centre (NOC) and Security Operations Centre (SOC) domains. The expert knowledge, along with statistical methods like correlation analysis, is used to create and select a set of features (e.g. mean and variance of some numerical value for a given time window or frequency of words in textual data) from the raw data, that are most meaningful for a given problem that the model aims to solve. Some features may be "synthetic," i.e. created from a combination of existing features. In the training phase, the prepared data is used to obtain model parameters, typically by minimizing a certain error function.

The model evaluation process serves to minimize generalization error, i.e. the error exhibited by the model when faced with data set that is different from the training data set. This is typically estimated using validation data sets, reserved within the existing data, and fed to the model post-training. The metrics used to measure this error depend on the nature of the parameter to be predicted. Since telco systems are considered critical infrastructure, a separate test bed may be required for the training and evaluation of any new model.

While real-time operational data may be used to train AI models, during the training and evaluation phase, these models are decoupled from downstream systems. For offline models, i.e. those with separate training and implementation phases, evaluation is performed entirely within the established test-bed. Online models, on the other hand, undergo continuous evaluation, both within, and outside the test environment.

5.2.3 Model deployment mechanisms

5.2.3.1 Introduction

Model deployment is often referred to as model distribution. To take advantage of the operators' Core Network (CN) computing resources, a model can be engineered and evaluated at the core network nodes or AI/ML Application Functions (AI/ML AF) outside of the operator's core network and deployed for execution to the User Equipment (UE) nodes.

There are two fundamental ways to distribute information between CN or AF and UE(s). They are utilizing the Control Plane (CP) and User Plane (UP). In addition, a hybrid distribution would utilize both CP and UP. Clauses 5.2.3.2, 5.2.3.3, and 5.2.3.4 discuss possible model distribution use cases.

5.2.3.2 Model distribution using CP

The main advantage of model distribution using CP is its confidentiality, integrity, and replay protection over the air while protected with NAS security. Another advantage is full operator's control over CP and NAS security. However, a potentially large size of transferred models can adversely affect CP availability for its main purpose of assuring signalling for mobile communications.

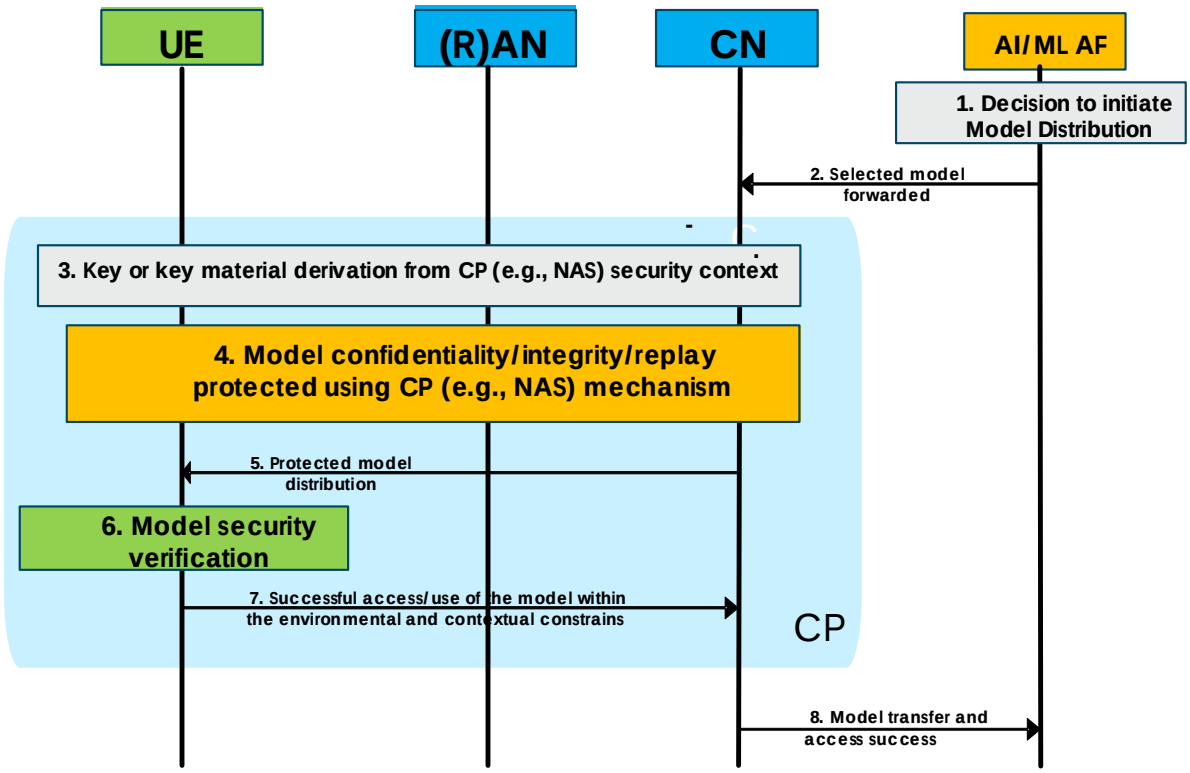


Figure 5.2.3.2-1: Illustration of the CP-based approach to the ML model distribution

In Figure 5.2.3.2-1, steps 2 and 8 are realized over the pre-provisioned secure communication channel between the operator's Core Network (CN) and the trusted AI/ML Application Function (AI/ML AF).

In step 3, the CN derives model distribution security context from the existing CN security context (e.g. NAS security context) and uses this security context for protection of ML model distribution from the edge of CN through RAN and into the UE.

5.2.3.3 Model distribution using UP

UP use for model distribution has the advantage of being able to accommodate large ML model sizes. UP might be used with Application Layer security (e.g. end-to-end) and as such can be easily outsourced by an operator. However, being opaque to the operator is one of the main disadvantages. The user QoE is mainly attributable to the serving mobile operator. At the same time, the model that is distributed over UP cannot be controlled by the operator. One of the examples of such discrepancy could be the case when the model that optimizes handovers is distributed from the entity that is not under the control of the operator using the means that are not under the operator's control (i.e. end-to-end confidentiality protected UP). However, when users' QoE is not satisfactory because of slow or failed handovers, it is the serving operator that will be blamed for poor QoE.

Various ML model ownership and custodial arrangements are also not fully supported when a model is distributed using UP. Figure 5.2.3.3-1 illustrates the UP-based approach to the ML model distribution.

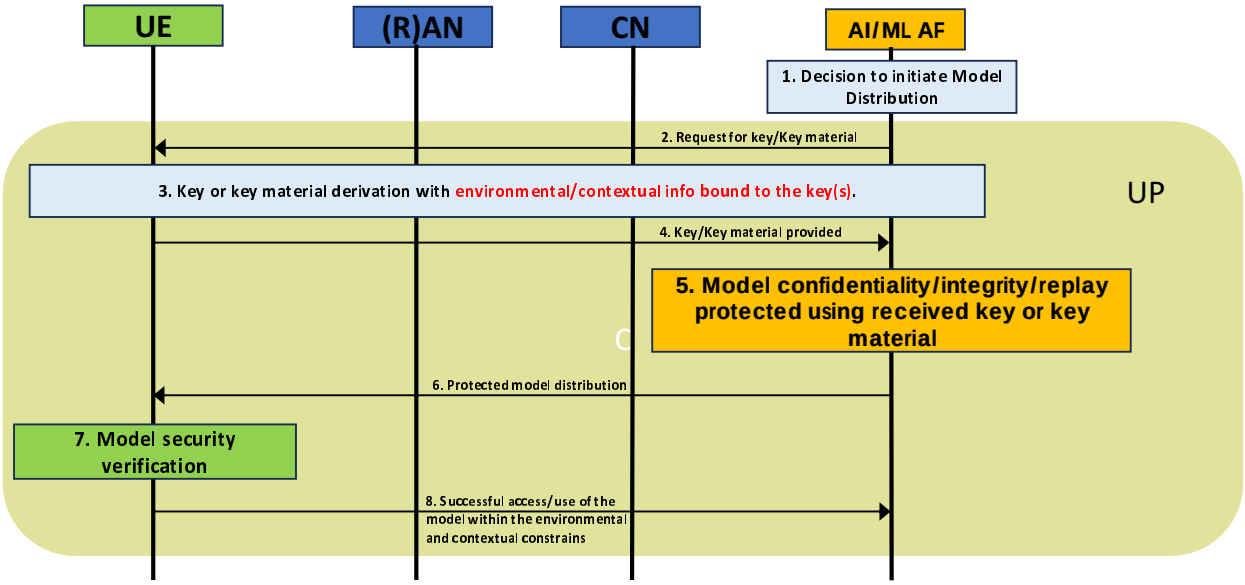


Figure 5.2.3.3-1: Illustration of the UP-based approach to the ML model distribution

5.2.3.4 Hybrid model distribution using CP and UP

Hybrid model distribution aims to remediate the CP-only and UP-only modes. Such a mode uses CP to set the cryptographic model distribution control that is shared between the home operator, serving operator, and model owner/custodian. After the CP-assisted framework for model transfer is set, the UP is utilized to distribute the ML model protected by the CP-assisted cryptographic framework.

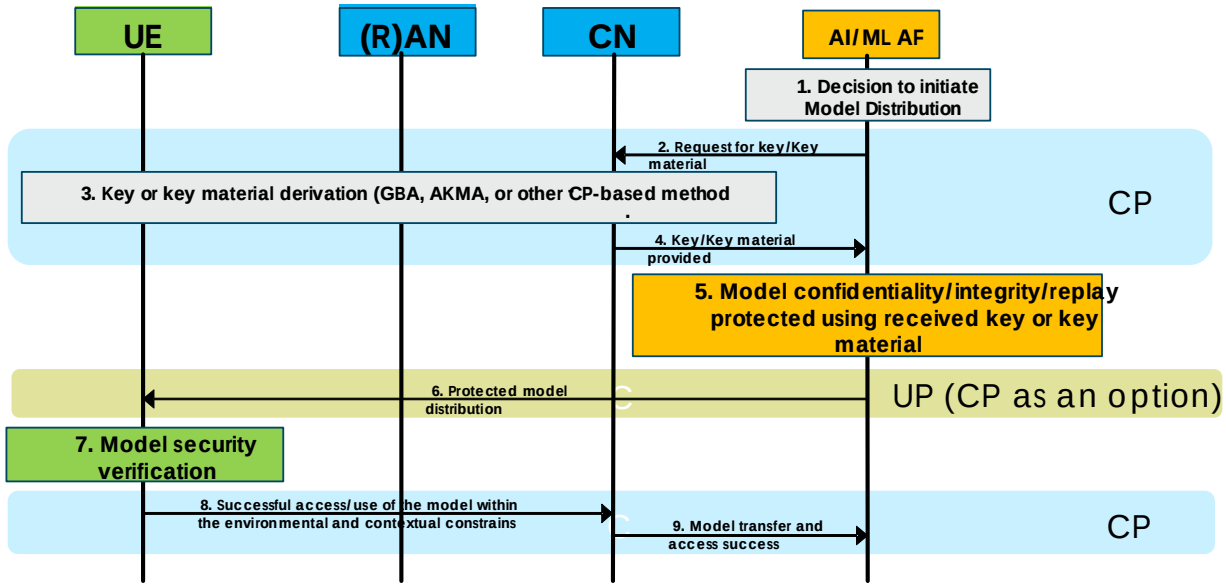


Figure 5.2.3.4-1: Illustration of the hybrid approach to the ML model distribution

5.2.4 Model operation (i.e. use of the ML model for prediction and inference) mechanisms

AI agents are software systems that use AI models along with interfaces to other software, hardware, humans, or other agents to pursue goals and complete tasks on behalf of users. AI agents may show reasoning, planning, and memory and have a level of autonomy to make decisions, learn, and adapt. Such agents may be deployed within numerous functions of a network, for e.g. triggering pre-defined automated workflows to redistribute traffic on prediction of increased congestion.

6 AI/ML use cases for telecom and their impact on AI security

6.1 System monitoring

6.1.1 Introduction

System monitoring could be seen as a prerequisite to gathering information on the status of the system to perform further investigations as described in the subsequent clauses. The data and information that is monitored may be on different layers in the system, i.e. in the general availability of resources. For example, such information may be Network Functions (NF) or load situations in terms of memory consumption and throughput. The monitoring can also span into protocol states, malformed packets (header values out of range, new header parameters, etc.) protocol failures, and the respective error codes on different protocols such as RRC, PDCP, NAS, SBI, etc., or on application layer in case operator native applications like SIP/IMS media are involved. Further information can be collected on a per NF basis or in a more fine-granular way on a per UE basis, depending on the point of data collection. The 5G system already is designed to some extent with some monitoring capabilities: the Network Data Analytics Function (NWDAF) [i.1] can collect data from various sources i.e. via subscription to events from AMF, SMF, PCF, UDM, NSACF, AF (directly or via NEF) and OAM, from Data Collection Coordination Function (DCCF), from Messaging Framework Adaptor Function (MFAF), from data repositories (e.g. UDR via UDM for subscriber-related information) and ADRF (Analytics Data Repository Function), or NRF for NF-related information. The NWDAF may contain one or two of the Analytics Logical Functions (AnLF) for deriving analytics information and the Model Training Logical Function (MTLF) for training machine learning models.

Further, the 5G system comprises the MDA capability [i.2] for mobile networks and services management and orchestration which can consume the analytics provided by the NWDAF. The MDA has system monitoring capabilities and can process and analyse a large number of different data related to network and service events and status. The data monitored by the MDA can be current or historic data, examples are performance measurements, KPIs, Trace data including MDT/RLF/RCEF reports, QoE reports, alarms, configuration data, network analytics data, and service experience data from AFs, etc.

The MDA provides its output by the MDAS (Management Data Analytics Service) producer to the consumer NFs, which contains analytics output in terms of statistics or predictions, root cause analysis issues, and, potentially, recommendations for enabling necessary actions for network and service operations.

From the Lifecycle Phase perspective, this clause deals with Day 2.

6.1.2 Anomaly detection

System monitoring is a prerequisite for Anomaly Detection and for developing the definition of the normal state and behaviour of the system to draw a comparison to events or behaviour that deviate from the normal state. There is some basic Anomaly Detection already defined within the scope of the network analytics and the management data analytics features. The NWDAF can identify a group of UEs or a specific UE with abnormal behaviour, where abnormal is defined as the UE being either hijacked or misused, running malicious applications on the UE. The NWDAF can provide analytics on the following two analytics types based on the exception identity parameters:

- Mobility-related analytics: Unexpected UE location, Ping-ponging across neighbouring cells, Unexpected wakeup, Unexpected radio link failures.

- Communication-related analytics: Unexpected long-live/large rate flows, Unexpected wakeup, Suspicion of DDoS attack, Wrong destination address, Too frequent Service Access.

Upon detection of abnormal behaviour based on the analytics of the exception identity parameters, the NWDAF informs the consumer e.g. the MDA, about the analytics results.

The MDA can utilize AI/ML technologies for the analytics and the MDA function may be deployed per individual MDA capabilities, the relevant ML entities are used for inference. Considering the vast amount of input parameters the MDA can collect from various sources, it can identify current issues in terms of network performance and services. The MDA has the capability of assisted fault management, i.e. it can monitor the status and current behaviour of various network functions and predict potential network failures in advance based on the running trends.

6.1.3 Root cause identification

The analytics of the MDA can already contain specific root cause identification of the detected issues. There are already several requirements on the MDA capabilities to provide a root cause identification e.g. network slice throughput issue(s), E2E latency issue, network slice load issues and recommended actions, energy efficiency issue, and mobility performance issues. Such detection and identification can be performed on both, raw operational metrics or text logs, generated by the deployed and instantiated NFs. To achieve this, LLMs or other autocorrelative models may be deployed for log-based detection, streamlined classification, root cause clustering, or regression purposes. A combination of these approaches may be used in an ensemble for greater detection performance.

6.1.4 Predictive maintenance

Information from the MDA pertaining to the operational characteristics of the underlying network infrastructure may be used as inputs to NN models trained to predict expected hardware failure times and flag components for pre-emptive replacement or repair. For example, a model may be used to ingest Self-Monitoring, Analysis, and Reporting Technology (SMART) metrics from Hard Disk Drive arrays, monitor their deterioration, and provide proactive drive replacement schedules.

6.2 Intelligent networks

6.2.1 Introduction

Traditionally, in the telecom context, Intelligent Network (IN) allows functionality to be distributed flexibly at a variety of nodes in and outside the network and allows the architecture to be modified to control the services. AI in telecom networking offers several key advantages that are transforming how networks are managed and operated. AI significantly boosts network efficiency by automating routine and complex tasks. This automation leads to faster resolution of issues, more efficient resource allocation, and reduced operational overhead.

In turn, high-performance networking is a critical component of the technology infrastructure that enables AI applications to operate efficiently and securely.

6.2.2 Ability to self-heal

Given an identified (e.g. by Root Cause Identification subsystem) issue, remediation is typically performed through some combination of manual actions performed by a human, along with the application of infrastructure configuration changes, and occasionally, software code patches. The actions that need to be performed manually by a human in order to rectify the issue may be expressed in the form of a set of remediation steps, a natural language construct. The set of all necessary steps, code patches, and configuration changes required to respond to a given issue may be termed a Course of Action (CoA) for the issue. All elements of a CoA can potentially be generated by an LLM that employs RAG, whose input is the identified issue. If the CoA elements consist entirely of configuration and code changes, it is possible for these changes to be incorporated automatically by the concerned systems, possibly through an intermediate orchestration system. This allows for limited self-healing of the system. Typically, it would be considered belonging to the Day 2 phase.

6.2.3 Root cause identification

The analytics of the MDA can already contain specific root cause identification of the detected issues. There are already several requirements on the MDA capabilities to provide a root cause identification e.g. network slice throughput issue(s), E2E latency issue, network slice load issues and recommended actions, energy efficiency issue, and mobility performance issues.

6.2.4 Predictive maintenance

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6.2.5 Dynamic optimization

6.2.5.1 Mobility Optimization

Mobility Optimization is one of the use cases studied in [i.3] and specified in [i.4] and happens in the operational phase, i.e. Day 2.

The use case aims to minimize performance loss due to unsuccessful or erroneous mobility management events. Mobility management is expected to guarantee the service-continuity during the mobility by minimizing call drops, Radio Link Failures (RLFs), unnecessary handovers, and ping-pong. In the future, it is expected that handovers will be increasing in numbers as the coverage of a single node decreases and UE mobility gets higher and more active. In addition, for the applications characterized by the stringent QoS requirements such as reliability, latency, etc., the Quality of Experience (QoE) is sensitive to the handover performance, so that mobility management should avoid unsuccessful handovers and reduce the latency during the handover procedures.

The RAN AI/ML framework studied in 3GPP TR 37.817 [i.3] and specified in RAN specifications ETSI TS 138 423 [i.4], includes several network entities exchanging AI/ML-related information for the purposes of data collection, data inference, output, and feedback. These network entities are UEs, RAN nodes, and potentially OAM nodes depending on the architecture. The RAN AI/ML framework specifies three use cases including Mobility Optimization for which the UEs and RAN nodes provide input and inference data and the RAN AI/ML framework on RAN and potentially OAM nodes provide output and feedback data to relevant nodes. Note that other use cases include Network Energy Saving and Load Balancing.

An OAM and /or NG-RAN node may train a model or perform inference using UE-related information acquired by the RAN node (e.g. UE location information and UE trajectory prediction), and the information obtained from neighbouring RAN nodes (e.g. UE mobility history information).

The RAN AI/ML framework includes information transfer procedures from UEs and RAN nodes. UE-related data are annotated with temporary UE identifiers or UE measurement identifiers. The information generated by UE and RAN nodes stays within the 3GPP network domain and is not exposed to third parties.

6.2.5.2 Load Balancing

The Load Balancing use case aims to distribute the load evenly among cells and areas of cells, to transfer part of the traffic from congested cells or congested areas of cells, or to offload users from one cell, cell area, carrier, or RAT to improve network performance. This can be done by means of optimization of handover parameters and handover actions. This activity is considered a typical Day 2 operation.

6.2.6 Automated network design

The following clause provides an example of an AI/ML pipeline dedicated to automated network design. It utilized the methodology and terminology that is defined in clause 3.1.1.

Network Requirements Analysis and Design tasks typically occur in the Day 0 phase. Requirements Gathering, Strategic Goal Setting, and Outcome Analysis consume a significant proportion of the efforts involved in network design, the output of which is typically captured as documents intended for a wide variety of stakeholders across executive, legal, business, and technical fields. LLMs are expected to be adopted extensively in this stage to perform tasks such as refinement of abstract strategic goals into clear business objectives, and subsequent transformation of these objectives into specialized technical requirements for the set of intended audiences.

System architecture specifications constructed using these technical requirements may then be fed to specialized generator or optimizer models to create technical designs for the system. Generative AI techniques can also be used to transform technical requirements into system architecture specifications and initial network layout plans, which can then be optimized using predictive AI/ML techniques. Low-level deployment plans typically involve the use of boilerplate templates requiring a degree of manual entry. Such tasks are likely to be offloaded to multimodal LLMs, i.e. models that can produce multiple forms of output, e.g. text and images, within a single query context.

Deployment configuration, considered a Day 1 task, can also potentially be aided through Generative AI. ICT infrastructure deployments increasingly make use of Infrastructure as Code (IaC) and virtualization techniques to accelerate the path to operational readiness. These techniques typically make use of configuration files to describe the intended architecture, topology, and resource allocation of application infrastructure as well as their behaviour. Such configuration files, hitherto written manually, are suitable targets for automation through LLM generation.

6.3 Managed telecom services

6.3.1 Use cases

The following subclauses add detailed descriptions for managed telecom services use cases. These use cases mostly relate to Day 2 in the network lifecycle described in clause 3.1.1.

6.3.2 Ticket (e.g. trouble ticket, CR) classification and routing

In many cases, the tickets contain data entered by humans in natural language such as a description of the problem and the steps already taken. While frequently some initial classification of the problem is provided by the person opening the ticket, it can also be marked as 'other' or be imprecise. In any case, the LLM can analyse the problem description text to decide where to route the ticket. For example, which support team should handle it (customer care, technical, etc.) or whether the escalation to the higher tier is already in order. Moreover, after closing the issue, the LLM can prepare a digest of the remedy and publish it in the form of a 'best-known procedure' for more widespread use (e.g. FAQ webpage). Furthermore, the knowledge of how to solve a given problem (or information that is not solvable) can be fed back to the AI model, leading to faster and more precise answers in the future.

6.3.3 Customer churn prediction

At a higher level, LLM-based sentiment analysis is also possible with the tickets (see clause 6.3.1) and other sources (e.g. the history of chats clients had with a company chatbot or social media posts). This can be combined with data obtained from network infrastructure measurements such as end-to-end delay or downtimes (see also clause 6.3.4), facilitating trend and changepoint analysis. This combined predictive and generative AI-based study may ultimately give an operator an early warning about possible customer churn.

6.3.4 Service (e.g. SLA) assurance

Service level assurance utilizes several elements mentioned previously, with monitoring and data collection being a starting point. AI/ML-based analysis of the data can predict, for example, a possible SLA violation (e.g. probability of exceeding an agreed end-to-end delay value within a given time frame is 90 %), analysing the cause of it (delay is not because of the network but due to CPU processing instead of GPU), and taking preventive action before the actual violation occurs (offload the processing from edge node that has no free GPU to a central cloud with free resources). In all of these steps, AI/ML models can be used for prediction, inference, Course-of-Action generation, and execution.

7 Vulnerabilities and mitigation strategies of AI/ML models deployed in telecom use cases

7.1 Attacks on System Monitoring

7.1.1 Introduction

It is envisioned that the compiled executable AI models will be used as AI Agents attached to network functions or form separate network functions themselves. AI agents that improperly or maliciously function may be potent attack vectors in telecom networks. Their instantiation and use need to be strictly monitored

Each use case identified in clause 6 of the present document is represented below with NIST mappings for the principal classification of adversarial AI technique possible and the attackers' goal, along with a brief commentary on the technique, references to implementations of such attacks, and possible steps for remediation or prevention.

The goal of the attacker targeting system monitoring can be either to directly disrupt the service or be more indirect like hiding further attacks or even exfiltrating data gathered by the monitoring system to only plan for the subsequent adversarial activities.

7.1.2 Anomaly detection

Most NN anomaly detection algorithms operate on time series data. Studies have demonstrated an attack on such models by inducing imperceptible perturbations into an input signal fed to the model [i.10]. This form of attack causes models to misclassify anomalous signals as normal operating behaviour with high confidence, propagating these errors to upstream decision-making systems. Dataset poisoning is also a viable attack strategy as demonstrated by Nguyen et al. Here, by introducing specific patterns of malicious activity into the training dataset as benign, they created models that were effective in their intended operation, but consistently ignored the intended malicious pattern. Log analysis models are also susceptible to perturbation attacks [i.11]. LLMs are further susceptible to indirect prompt injection, where logs (or traces) to be processed may contain malicious code or instructions that can then be executed by downstream systems.

7.1.3 Root cause identification

Targeting the MDA system can offer an attacker the opportunity to induce misdiagnosis of a problem root cause, thereby either triggering an erroneous response or distracting response teams to mask a secondary attack. The specific techniques that may be employed by an attacker depend on the RC identification mechanisms used. Models relying on stochastic techniques such as AI/ML models may be poisoned during the training phase, either by means of backdoor injection, or dataset tampering. Prevention of such attacks lies primarily in the training phase, where models will need to be trained with adversarial samples to improve robustness.

7.1.4 Predictive maintenance

Similar to the case for stochastic MDA systems, NN models trained for predicting system failures may be compromised by poisoning or signal tampering. Time series models, which are typically employed for such monitoring are also particularly susceptible to perturbation attacks, whereby an attacker introduces specific, smooth perturbations in input signals [i.10] that cause misclassification errors. Such attacks may be used both for generating false positive failure events, that may be used to distract from secondary attacks, and false negatives, that will induce system failure in the future. Training robustness techniques and ensembling may be used to alleviate this risk.

7.2 Attacks on Intelligent Networks

7.2.1 Introduction

During the runtime phase, an attack on the AI-powered network control plane can allow an adversary to disrupt the service in several ways. These include limiting the system's ability to correctly execute the reconciliation loop or tricking it into making incorrect decisions regarding resource management. Also, when AI is used in the design phase, the attacker can exfiltrate data related to the planning or try to plant the backdoors that - after the deployment - will allow for executing further attacks.

7.2.2 Ability to self-heal

Poisoning of the training data in the case of CoA identification can lead to a set of actions that ostensibly address a given issue in the system but leave unaddressed an exploitable weakness. In the case of CoA execution, the employed LLMs may be induced to generate code with vulnerabilities. This is possible even in the case of unaltered publicly available models and can be exacerbated through training data poisoning to increase the likelihood of vulnerable code being produced for benign prompts. Sandboxing and malicious prompt detection may be used to minimize the effects of exploits inadvertently generated by the LLMs.

7.2.3 Dynamic optimization

The time series nature of network performance optimization presents the risk of perturbation attacks similar to those possible on predictive maintenance systems. When applied to the mobility optimization use case, it would then be possible for an attacker to artificially degrade user experience in specific cells by triggering handovers to sub-optimal cells. Load balancing may be disrupted by poisoning attacks that use a specific trigger to force the model to overload specific cells. The mitigation of these attacks follows the strategy outlined for time series models.

7.2.4 Automated Network Design

A potential for data extraction exists if an attacker is able to capture prompts given to the model, especially in the case that the model is provided as a public cloud SaaS. By repeating the prompts an attacker can recreate the output, or potentially expose confidential information fed to the model. This attack may be trivially undertaken, with the conclusion that prompts are not secrets. Such attacks can be mitigated to an extent by sanitizing prompts.

An attacker may attempt to manipulate the data being ingested by the model in order to induce incorrect results that may be propagated to higher-level systems or models, resulting in suboptimal designs, wasted resources, and potentially lost revenue.

An attacker may induce the model to generate insecure or weak code that may be exploited in the future, and have it enter the CI/CD pipeline of the target. This may be achieved by poisoning the datasets that the model was trained on, or by prompt engineering to create functional code with obfuscated vulnerabilities. It is possible to build such poisoned models that ostensibly pass safety tests.

7.3 Attacks on Managed Telecom Services

7.3.0 Introduction

A successful attack on customer-related services can damage the operator's reputation and potentially escalate to both monetary and legal consequences. This can occur if there is a violation of contractual agreements or if a leak of personally identifiable information is discovered.

7.3.1 Ticket (e.g. trouble ticket, CR) classification and routing

LLM models employed for ticket classification, especially those that are exposed to customer-originated tickets, are at high risk for prompt injection with the aims either of inducing privacy compromise or triggering hallucinatory behaviour. The latter represents a legal risk for the operator as, depending on the jurisdiction, LLM-generated responses presented to a customer are treated as on par with official documentation. Prompt sanitization and the use of RAG techniques for material such as FAQs presented to the customer can largely mitigate these risks.

7.3.2 Customer churn prediction

Time series attack techniques such as smooth perturbation may be employed by an attacker to induce churn prediction models to present an overly optimistic estimate, thereby causing revenue loss. The mitigation of these attacks follows the strategy outlined for time series models.

7.3.3 Service (e.g. SLA) assurance

Due to its compound nature consisting of sets of systems such as RCA, CoA generation, and CoA execution, the possible attacks, and the corresponding mitigation strategies, on SLA assurance systems may be considered as the union of the attacks and mitigations of their constituent systems.

Annex A: Bibliography

- ETSI TR 104 221: "Securing Artificial Intelligence (SAI); Problem Statement".
- ETSI TR 104 222: "Securing Artificial Intelligence; Mitigation Strategy Report".
- 3GPP TR 33.877: "Study on the security aspects of Artificial Intelligence (AI)/Machine Learning (ML) for the Next Generation Radio Access Network (NG-RAN)".
- 3GPP TR 33.898: "Study on security and privacy of Artificial Intelligence/Machine Learning (AI/ML)-based services and applications in 5G".
- 3GPP TR 33.738: "Study on security aspects of enablers for Network Automation for 5G - phase 3".
- ETSI TR 103 305-1: "Cyber Security (CYBER); Critical Security Controls for Effective Cyber Defence; Part 1: The Critical Security Controls".

Annex B:
Change history

Date	Version	Information about changes
9-2023	0001	SAI(23)18a016r1 - CR for clauses 2.2, 3.3, and 6.1 of SAI-0014 approved at SAI-18c
9-2023	0002	SAI(23)019008r1 - SAI-0014 - New sub-clause on GenAI vs traditional AI approved with editorial modifications at SAI#19
11-2023	0003	SAI(23)019b001 - Additional content for SAI-0014 approved at SAI#19b
12-2023	0004	Converted to the TR format as DTR/SAI-0011
12-2023	0004	SAI(23)001022 Additional content for clause 5.2 - ML functionality in telecom providers' networks. Approved at SAI#1 (SAI-TC)
01-2024	001	Converted from SAI-0014 (GR) into SAI-0011 (TR)

History

Document history		
V1.1.1	June 2025	Publication